WP6: Digital Twins Use Case

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About me: Eladio Montero-Porras

- Working in ULB-MLG as a Postdoctoral researcher
  - TORRES Project: 2025
  - WEL-T DT Project: 2026
- PhD obtained in 2024 at the VUB AI Lab
- Interests: complex systems, cooperation, artificial intelligence, digital twins

MOBILITY CHALLENGES

The Brussels Times MAGAZINE

A light at the end of the Brussels tunnel?

Brussels is quietly living with a split personality. Above ground, the city is slowing traffic, adding bike lanes and reclaiming space for people. Below ground, a crumbling 12-kilometre web of 1950s road tunnels still funnels a quarter of a million cars a day through the capital. As costs soar and politics stall, is it worth fixing the tunnels — or whether some should disappear altogether.

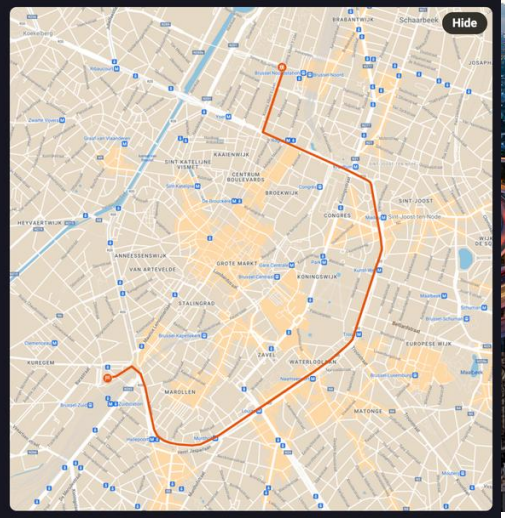
Monday, 23 February 2026

By Derek Blyth

Info-trafic Bruxelles Mobilité
@trafic
2h

Ce jeudi 12 mars: #manifestation à #Bruxelles. Perturbations de circulation attendues sur l'ensemble de la Région de Bruxelles-Capitale. Il est déconseillé de se rendre à Bruxelles en voiture ce jour-là.

L'itinéraire: Gare du Nord > Bd Roi Albert II > Petite Ceinture (R20) > Gare du Midi.



McKinsey & Company

Travel, Logistics & Infrastructure

Infrastructure technologies: Challenges and solutions for smart mobility in urban areas

Smart mobility solutions can transform city living and improve quality of life—but are difficult to implement at scale. A collaborative, ecosystem approach could be ahead.

a collaborative effort by Riccardo Boin, Timo Möller, Vadim Pokotilo, Andrea Ricotti, and Nicola senting views from McKinsey's Travel, Logistics, and Infrastructure practice and the McKinsey ure Mobility.

ACCUEIL • ÉCONOMIE • MOBILITÉ

De nombreux embarras de circulation à prévoir sur le Ring de Bruxelles dès le 1er août

Dès le 1er aout une large portion du Ring de Bruxelles sera en travaux. Ensuite, dès le 17 août c'est au niveau du carrefour Léonard que des perturbations sont attendues.

BELGA.

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DT USE CASE: TRAFFIC TWIN

- **Objective: evaluate the impact of road works in vehicular traffic using AI**
- **Traffic simulation:** allows to test multiple scenarios before deploying
- **Valuable tool for decision making in complex environments**
- **Administration Representatives:**
  - Paradigm
  - Brussels Mobility
  - Parking Brussels
  - Municipality of Ixelles

ROAD DEVIATION PLANS

- Deviation plans for roadworks are defined by domain experts
- In the Brussels area, many decisions on the implementation of deviation plans <u>are often not assisted by software tools</u>
- **No simulation tool for evaluating the impact of deviation plans**
- Brussels Mobilité is responsible for managing and implementing road deviation plans
- Police approves and/or can modify deviation plans

EXAMPLE: ROAD DEVIATION

Roadwork plan in Rue Scarron (Ixelles), as provided by Brussels Mobility

BRUSSELS MOBILITY
BRUSSELS REGIONAL PUBLIC SERVICE

Ixelles Elsene

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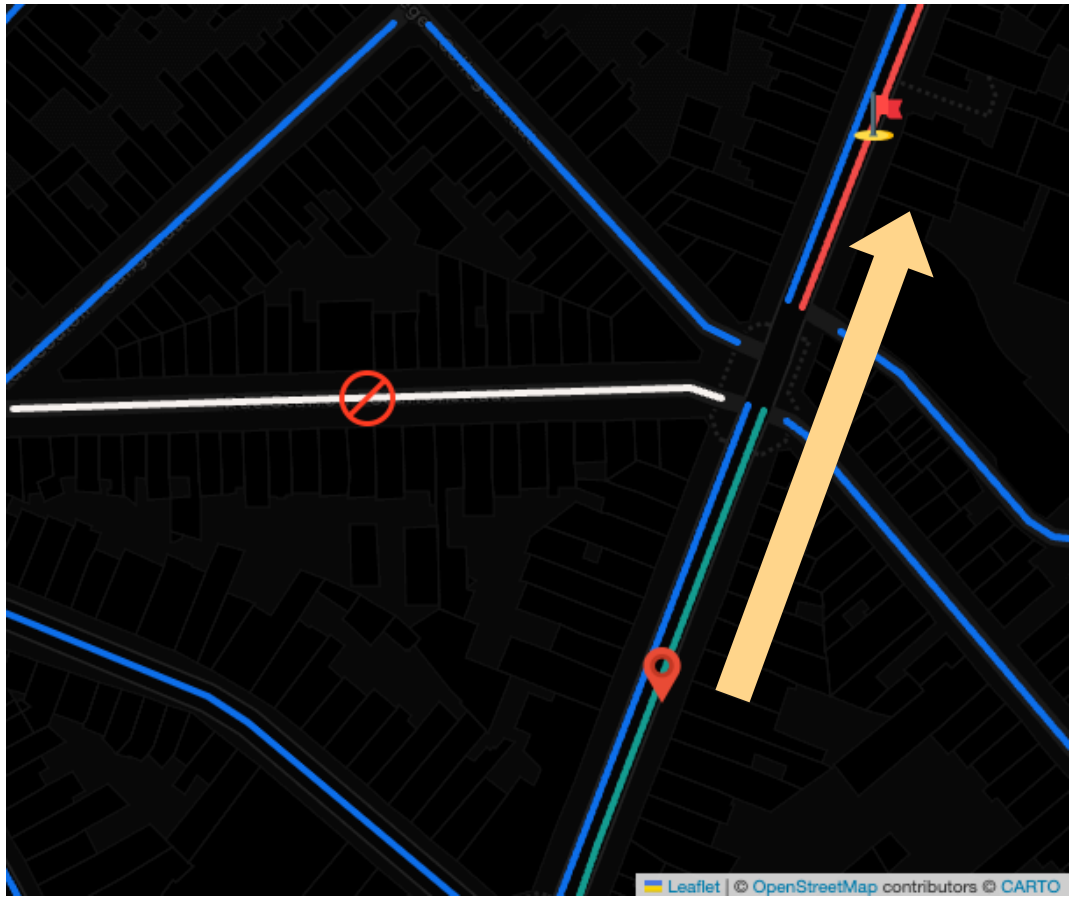
EXAMPLE: ROAD DEVIATION

Implementation of the deviation plan of Rue Scarron in TrafficTwin. The traffic is deviated according to the indicated flow



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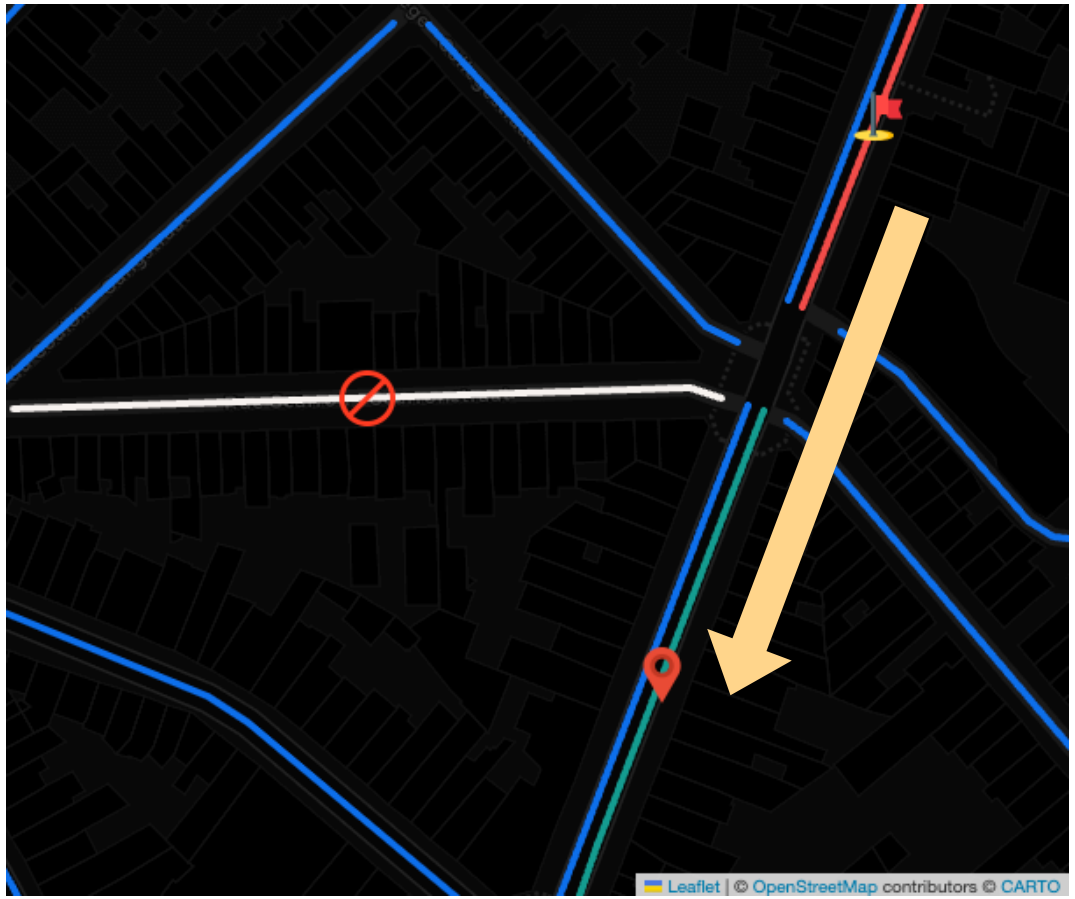

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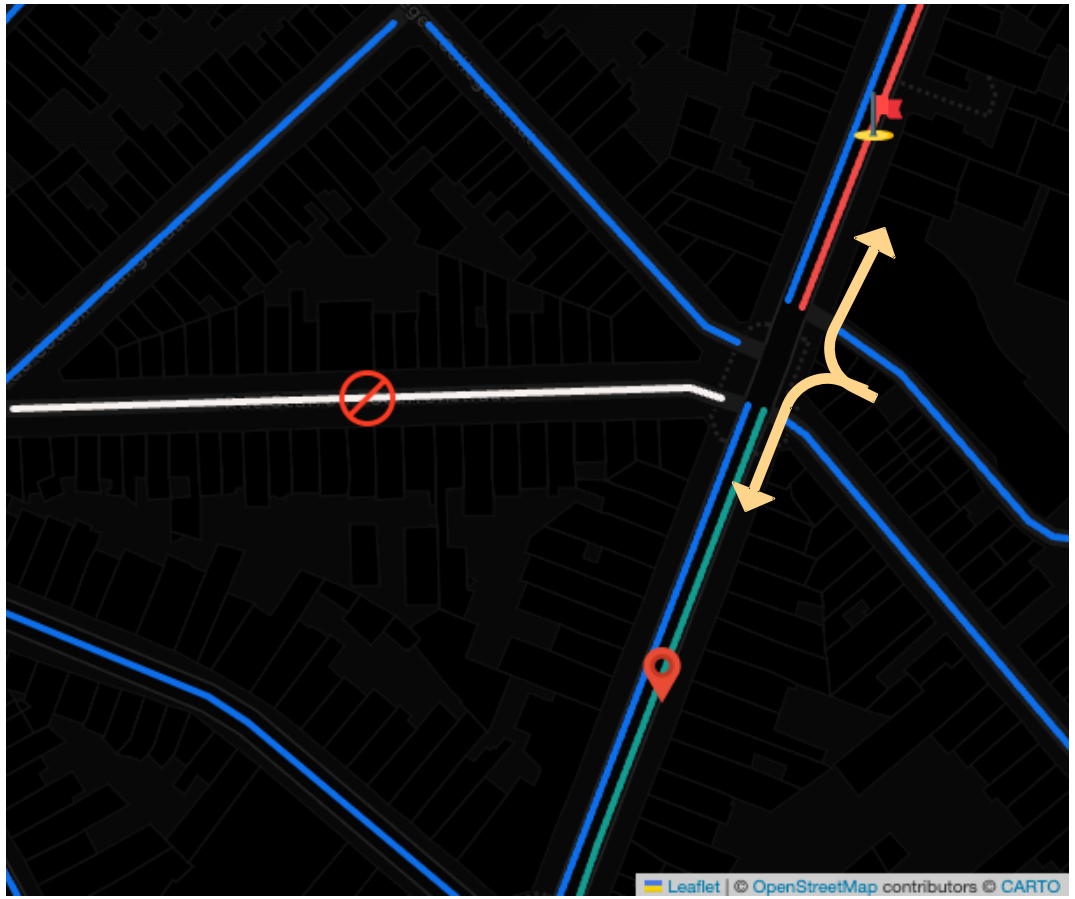

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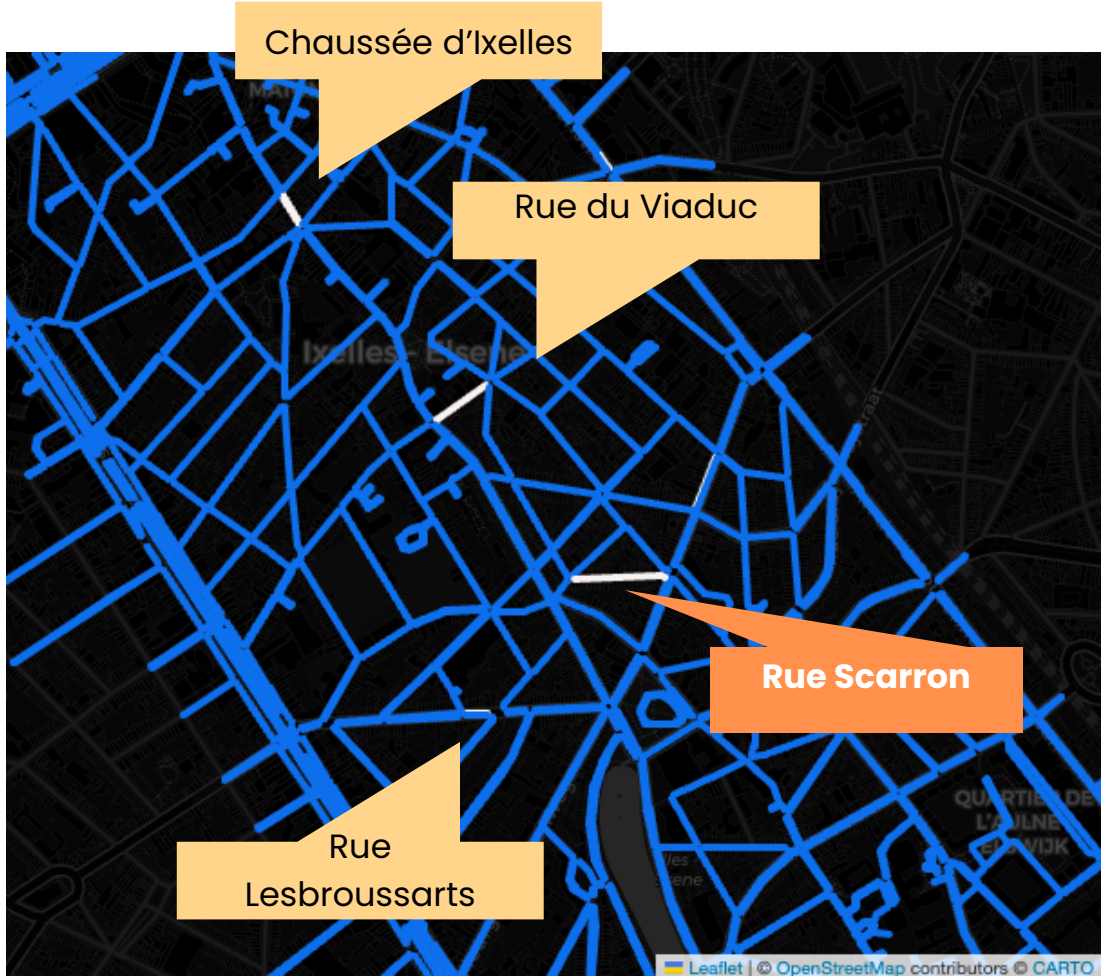
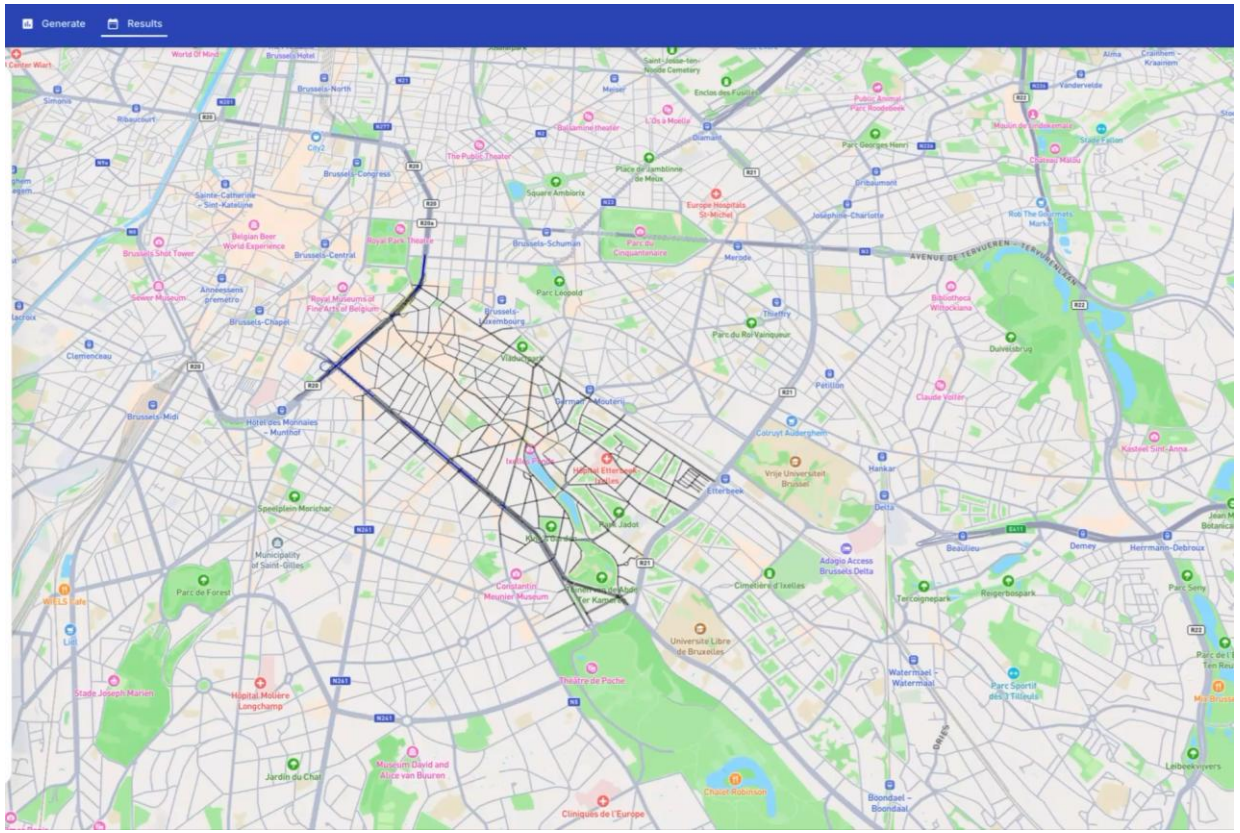
EXAMPLE: ROAD DEVIATION

Assuming the roadworks in Rue Scarron must be implemented, **what is the cost of other roadworks implemented at the same time?**



TRAFFICTWIN TOOL

1. **Interactive creation of disruptive events** by applying virtual road closures,
2. **Definition of alternative what-if scenarios** using different deviation plans,
3. **Simulation of these scenarios** using synthetic traffic models,
4. **Quantitative assessment** of each scenario,
5. **Interactive dashboards** to compare scenario outcomes.



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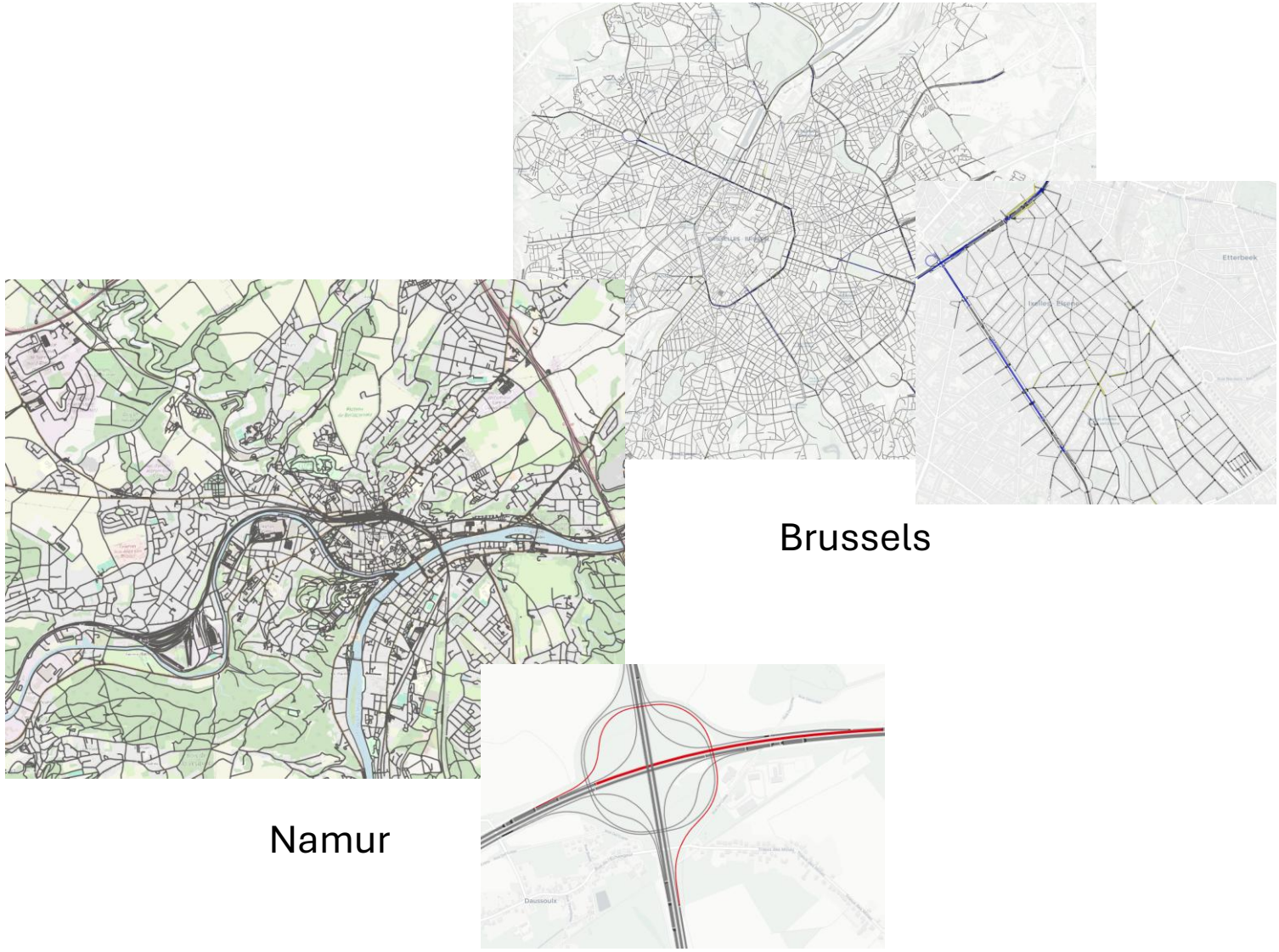
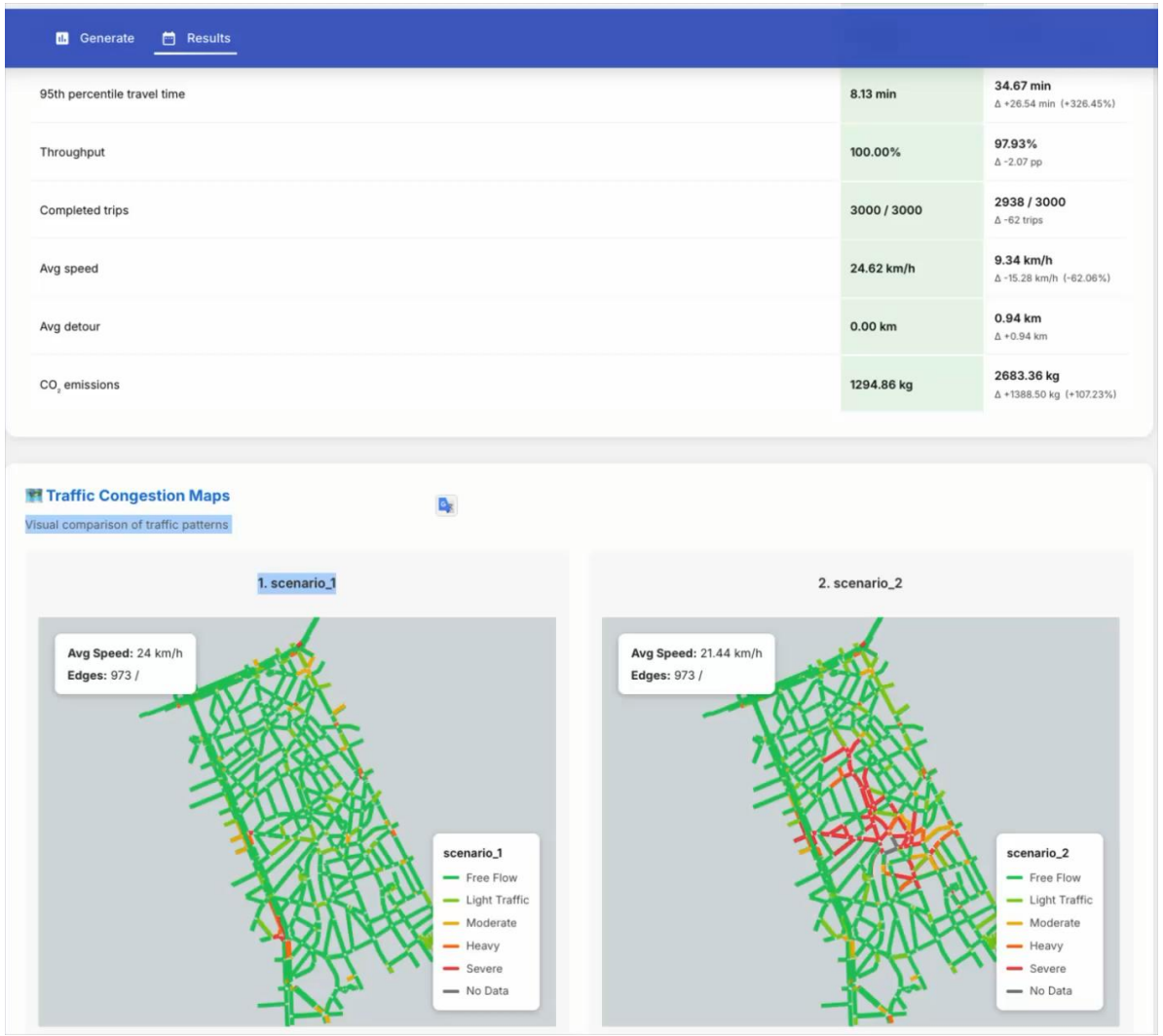
TRAFFICTWIN TOOL

- Uploading custom networks:
  - Allows for detailed analysis or macro simulations
- Creation of deviation plans
- Simulating using synthetic data or calibrated models



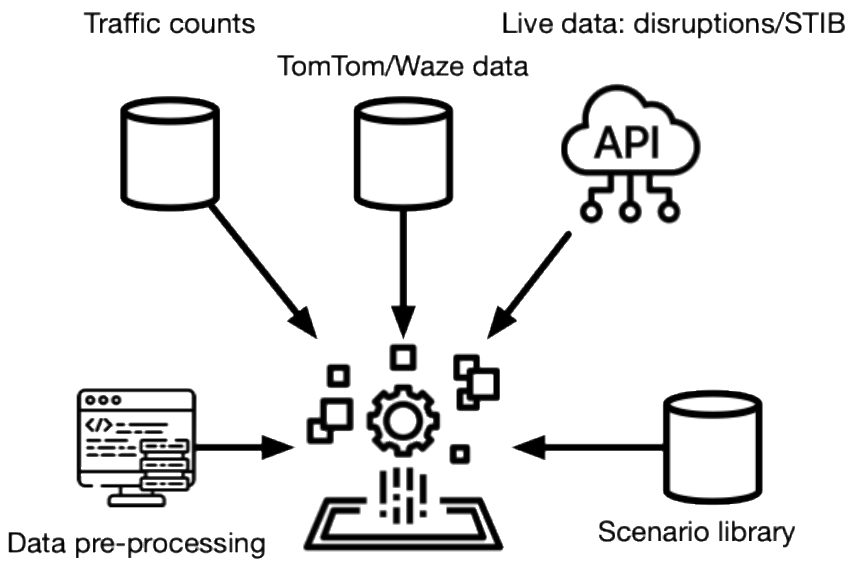
TRAFFICTWIN TOOL

- Scenario evaluation and comparison
  - Metrics: avg. speed, travel time, etc.
- Visualisation of the impact on the network



Future work: 6 months

- Crucial to integrate all different sources of sensor/gps data, etc
  - Different formats, times, spatial
- These data constitute high-fidelity data:
  - Scarce, expensive, in the past, but **realistic**
- Low-fidelity data: obtained from traffic simulations
  - Model-driven scenarios: events, accidents, etc.
  - These datasets are big, cheap, less realistic but they can simulate **events that haven't happened yet**.



Future work: 12 months

- Develop **explainable methods** that produce evaluations under uncertainty of closure options using calibrated traffic models, scenario generation using SUMO.
- This way, we can generate strategies for disruptions **before they happen.**
- **Scientific problem:** there is an infinite number of strategies for a given event, processing time is a challenge for real time usability.
- The final goal is that for a given event, we can derive a strategy that combines real data with synthetic scenarios.
- Potential use-cases: tunnel closures, events, protests, changes in urban infrastructure, rat running prevention, etc.

Development with the support of

- TORRES (2023-2025) Innoviris.
  - Monitoring and analysis of traffic data at the scale of a large city. Traffic Twin.
- TULIPE (2024) FARI.
- TRAIL–ARIAC (2021-2027) Digital Wallonia4.AI initiative.
  - Digital twins, and transfer learning to manage evolving data in dynamic systems, using causal analysis to uncover their fundamental mechanics.
- WEL-T (2026 -2029) FNRS/WEL-RI
  - Enhancing DIgital TwinS with scalable and interpretable AI
- POC (2026) SPW
- Proof of Concept: Traffic Twin